MANCHESTER 1824
The University of Manchester

EART11100

How to Build a Planet

Week 10
Weathering Processes

In week 4 we introduced the idea that the particles that make up sedimentary rocks come from weathering older sedimentary, metamorphic and igneous rocks.

But what are the processes responsible for weathering?

Weathering results from both chemical and mechanical processes.
These slides summarises the main processes of each.

What is weathering?

Water can break rocks and minerals down chemically and mechanically

Distinctive properties of water as a chemical weathering agent:

- Solvent
- Water is made up of bonded H^+ and OH^- ions
- Pure water is neutral (pH of 7) because it has equal numbers of H^+ and OH^- ions
- But rainwater is acidic (pH <7) because of reaction with atmospheric CO_2:

  Dissolution of CO_2 in water forms carbonic acid

  $H_2O + CO_2 \leftrightarrow H_2CO_3$

  which subsequently dissociates to produce hydrogen ions and carbonate ions

  $H_2O + CO_2 \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^-$
- Increase in H+ ions relative to OH- ions makes water more acidic and more aggressive dissolution agent

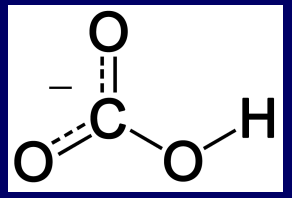
Major chemical weathering processes I

Simple solution (*congruent dissolution*)

Some minerals will dissolve altogether (e.g. calcite) liberating alkali- and alkali earth elements (Ca, K, Na, Mg) as soluble products (congruent)

$$CaCO_3 + CO_2 + H_2O \rightarrow Ca^{2+} + 2HCO_3^-$$

calcium



bicarbonate

Soluble calcium bicarbonate

Major chemical weathering processes II

Hydrolysis (*incongruent dissolution*)

In other cases dissolution is incomplete

H+ can replace metal ion in some silicate minerals by a process called hydrolysis

Result:

- metal ion released into solution
- secondary minerals are formed

Potassium alkali metal

Silicic acid

e.g. orthoclase feldspar - kaolinite (clay mineral):

$$2KAlSi_3O_8 + 2H^+ + 9H_2O \rightarrow 2K^+ + H_4Al_2Si_2O_9 + 4H_4SiO_4$$

A word of warning about clay minerals

- A confusing name!
- Clay is used for a grain size (<0.004 mm) and a group of sheet silicate minerals
- Clay minerals are usually, but not always, clay-sized
- Clay-sized grains are not all clay minerals. e.g. the quartz produced by weathering reactions is often clay-sized too.

Major chemical weathering processes III

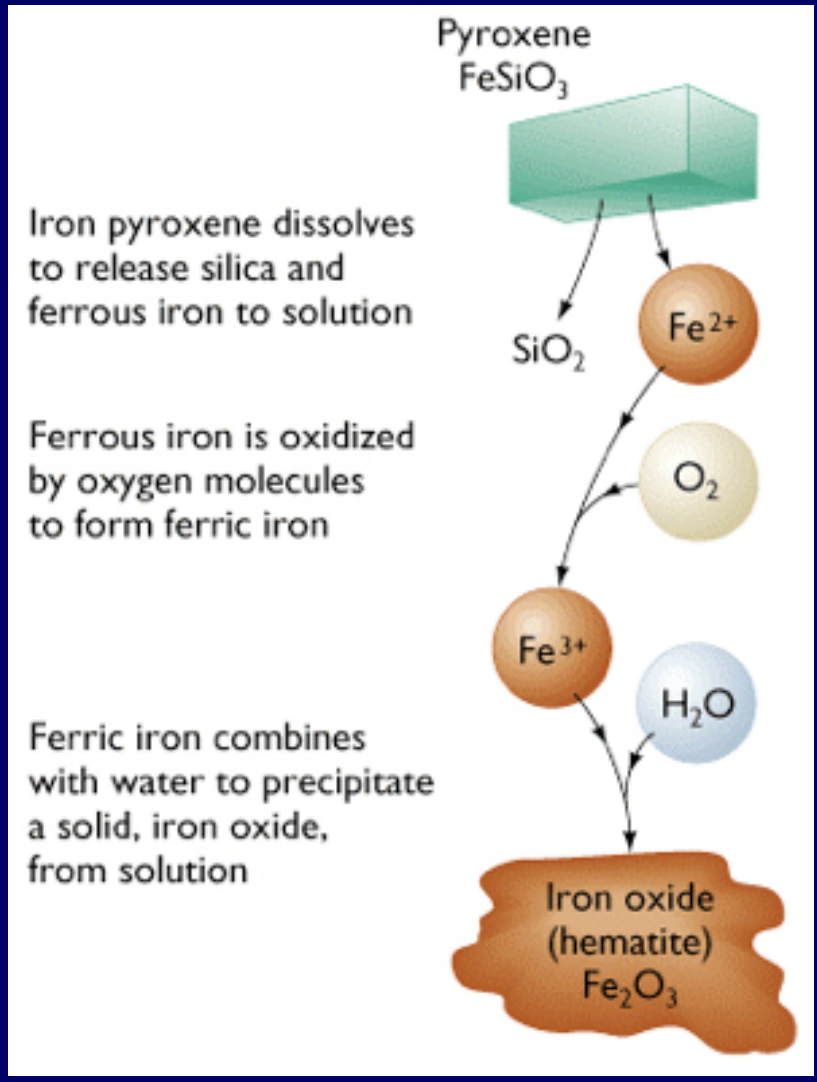
Oxidation

Oxygen dissolved in water can cause chemical alteration of iron, sulfur and manganese in silicate minerals

Oxidation results in loss of an electron from an element in a mineral, resulting in the formation of oxides



A rusty garden seat caused by oxidation of the iron metal



Example

Case study: weathering of a granodiorite

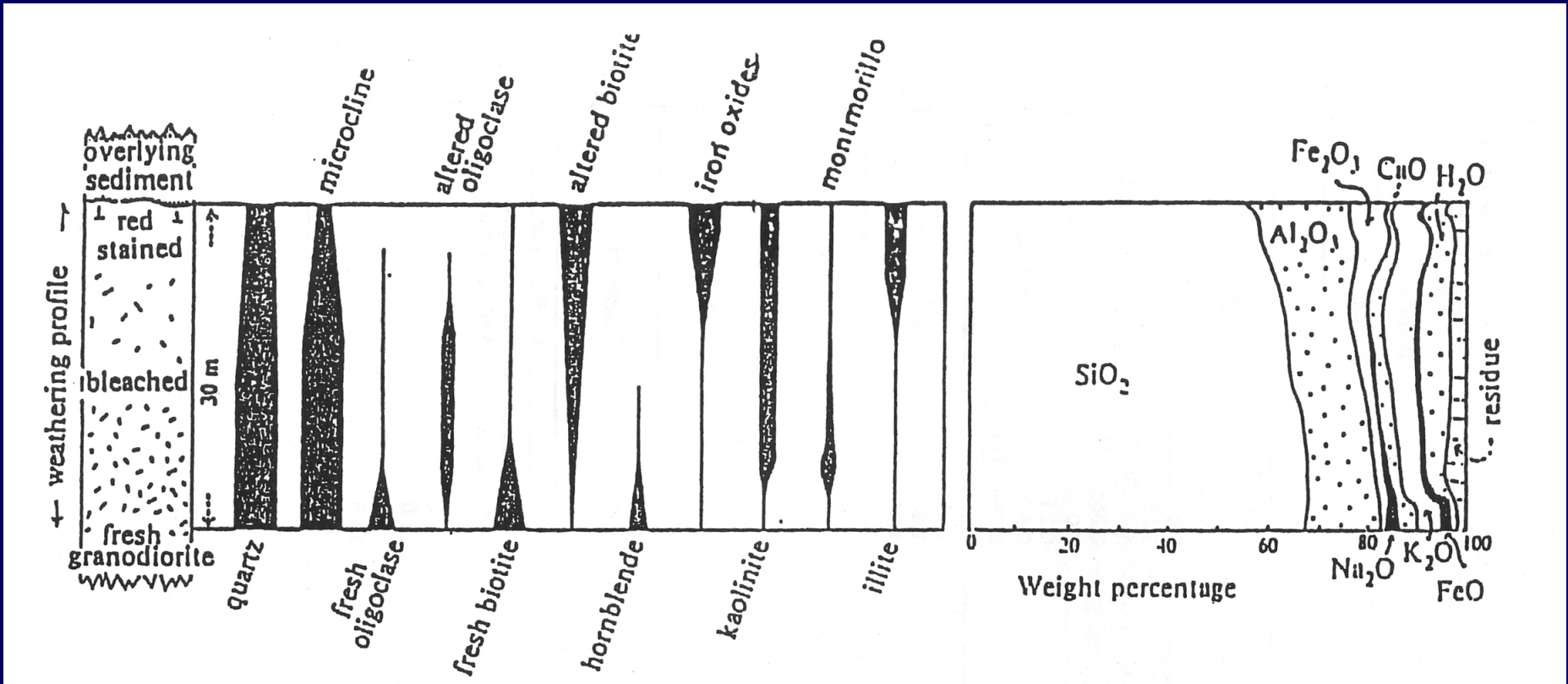
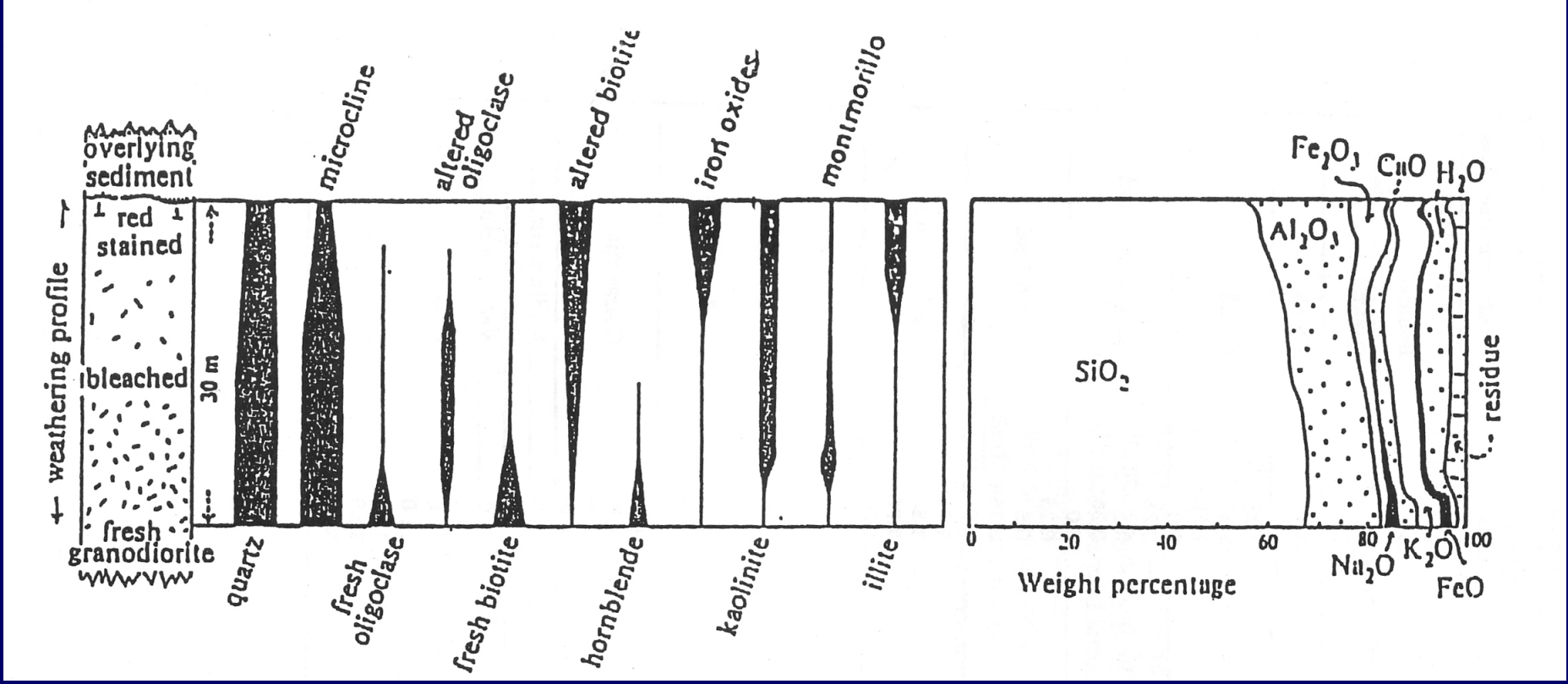


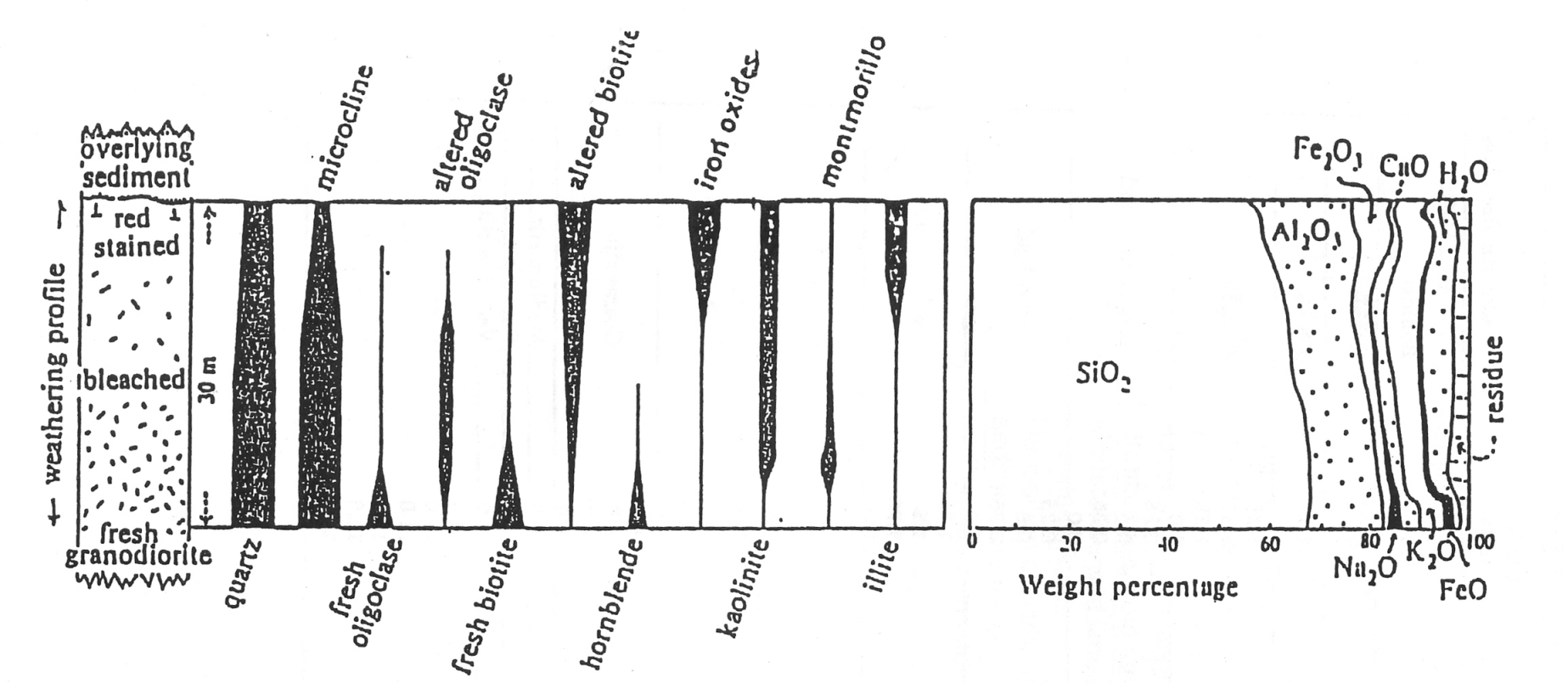
Figure shows section through weathered granodiorite (an igneous rock)

Let's look at a section through a granodiorite with depth from the ground surface to see how chemical weathering affects the mineralogy and chemistry of a rock

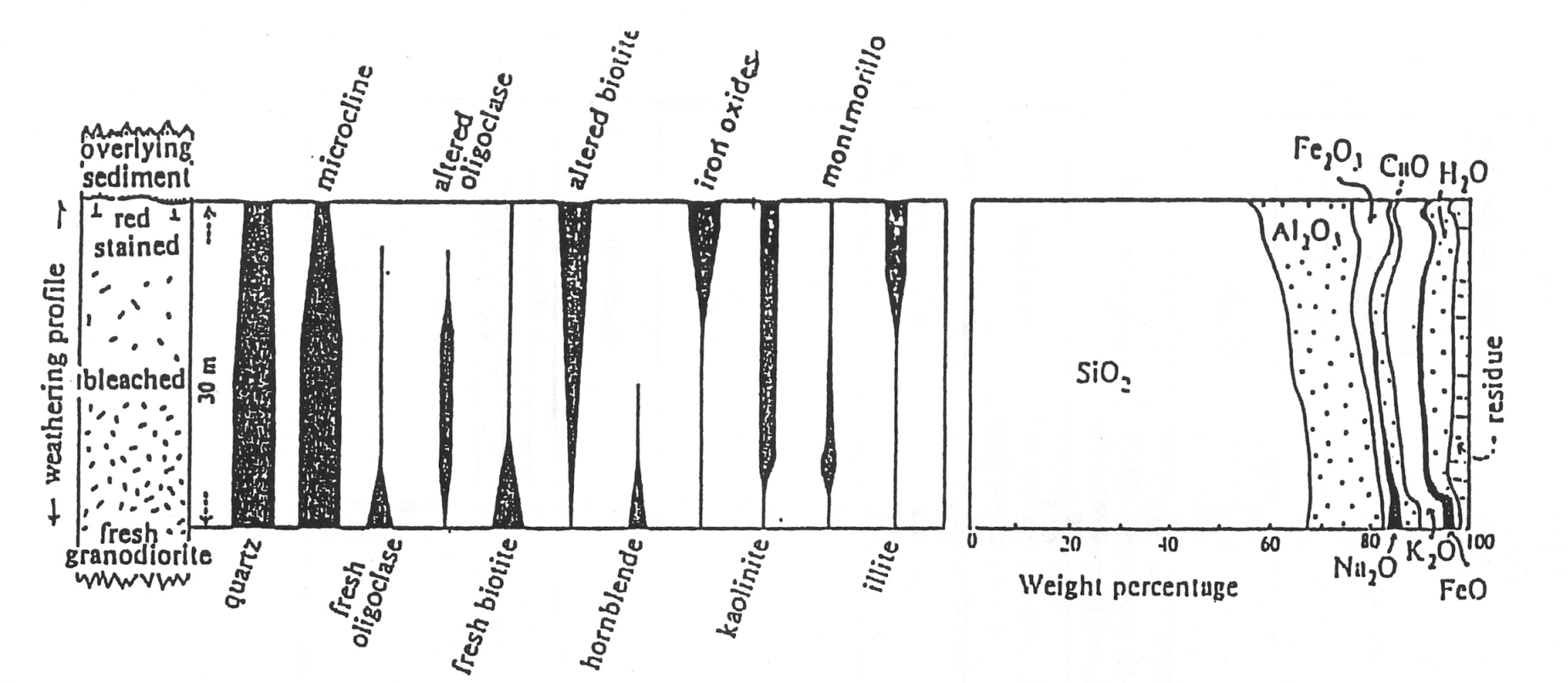


Fresh rock (found at depths of 25-30m below surface) has mineralogy:

- Quartz + Microcline alkali feldspar + Oligoclase plagioclase + Biotite + Hornblende
- Effects of weathering become more pronounced towards surface



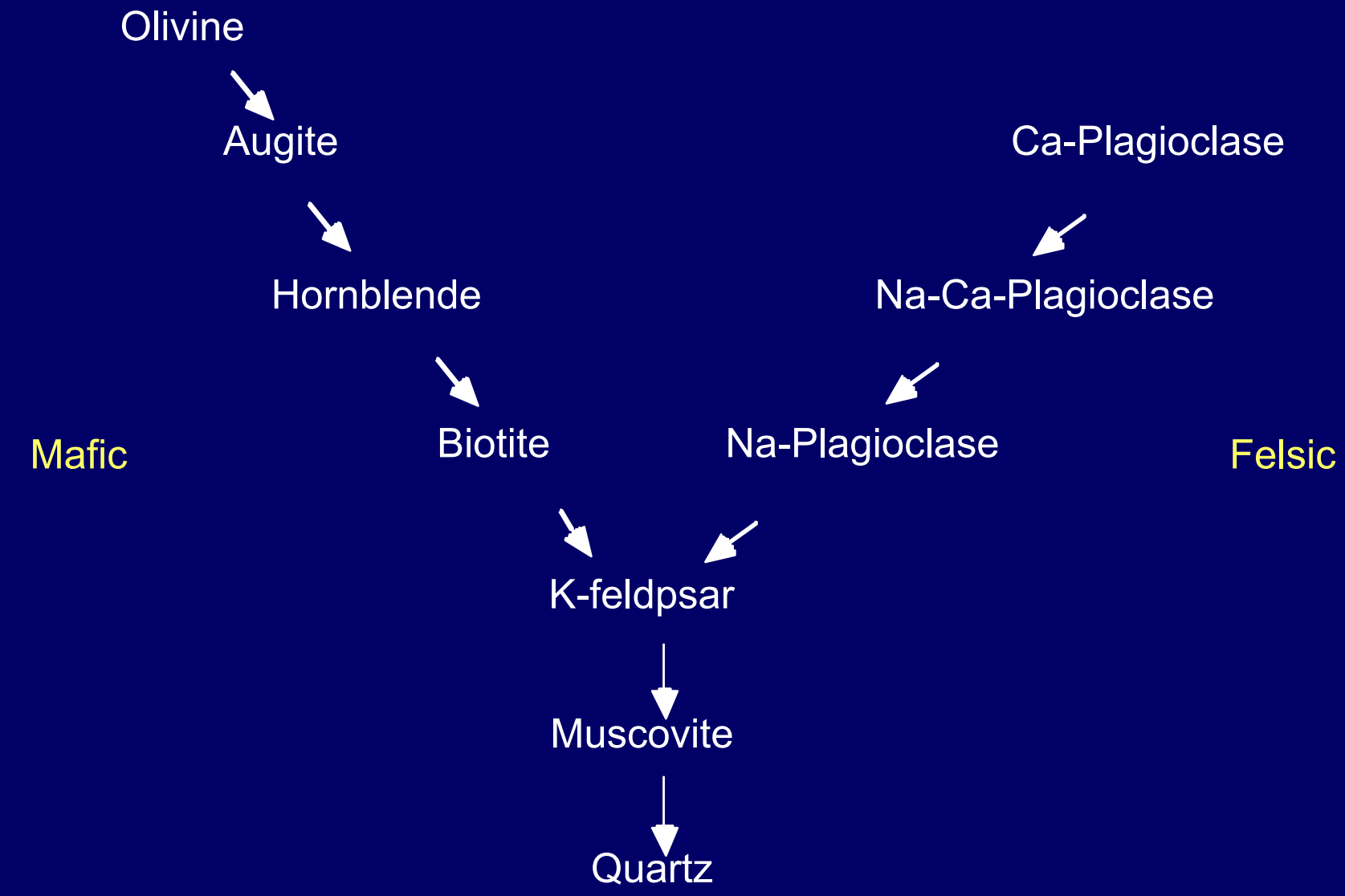
- Note stability of quartz
- Reasonable stability of alkali feldspar
- Instability of plagioclase feldspar, biotite and hornblende
- Addition of new minerals formed during weathering:
- Iron oxides
- Clay minerals



Weathering leaves some elements behind and carries others away in solution:

- Overall residual weathered rock becomes enriched in Aluminium, Ferric Iron and Potassium compared to parent granodiorite
- But depleted relatively in Silicon, Calcium, Sodium and Magnesium

Weathering series



Note that most stable minerals during weathering are those precipitated at lowest temperatures

Stability series is reverse of order of crystallization of minerals from magma

Mechanical / Physical Weathering

Process by which rocks are broken into smaller fragments, but without significant change in chemical or mineralogical composition

Major mechanisms:

- *Freeze / thaw*
- *Salt weathering*
- *Diurnal heating and cooling*
- Biological activity (e.g. *Plant roots expanding)*
- *Stress-Release*
- *River incision*
- *Glacial activity*
- *Wave activity*







Controls on weathering and erosion rates

Climate is main control on weathering rates

- Chemical reactions more rapid at higher temperatures

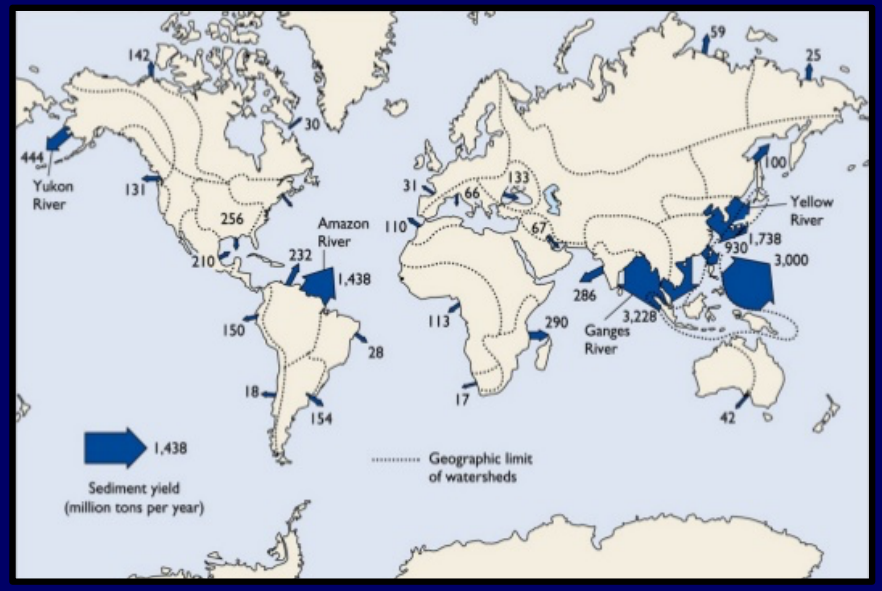
- Humid Tropics: chemical weathering is most rapid: intense leaching of both silicates and carbonate

- Temperate climates: chemical weathering less rapid but freeze/thaw becomes important particularly in upland areas, so mechanical weathering dominates

- Arid climates (hot or cold): weathering rates are low

What happens to the residual sediment?





Much fine-grained sediment is deposited at river mouths like here in the Ganges delta, India and Bangladesh

Today, 40,000 km^3 water runoff Earth's continents each year

One volume of ocean water recycled via the rivers every 40k yrs

Rates of erosion is variable:

- high in low latitudes where there are mountainous source areas
- moderate rates in temperate areas
- low in polar regions and hot deserts